III Semester B.Tech., (Branch Name)

Data Analytics – CES2103

INTERNAL ASSESSMENT – MINI PROJECT REPORT

School of Computer Engineering
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STATISTICAL ANALYSIS OF NASA CONFIRMED EXOPLANETS

A Data Analytics Mini Project

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1. Introduction

Exoplanets orbit stars beyond our Solar System, and their physical properties—radius, mass, orbital period, temperature and host-star spectral class—help compare planetary architectures. In this project, we analyse nasa_exoplanet_intelligence.csv (6,150 confirmed exoplanets across 31 columns) using descriptive statistics, visualisations, confirmatory hypothesis tests, multiple linear regression for planet radius plus a separate regression for missing equilibrium temperatures, KMeans/PCA clustering and a comparison of temperature-based versus stellar-flux habitable zones.

2. Objectives

- Assess data quality and preprocess missing or "Unknown" values while preserving observed measurements and storing regression estimates separately.
- Perform descriptive statistics, visual EDA, log10 transformations and Pearson correlation.
- Use Welch's t-test, chi-square tests, one-way ANOVA and two-way ANOVA to test hypotheses at $\alpha = 0.05$.
- Evaluate multiple linear regression using R^2, RMSE and MAE; apply KMeans and PCA for exploratory grouping.
- Compare the dataset's temperature-based habitable-zone flag with a luminosity/flux-based estimate.

3. Problem Statement

The archive combines heterogeneous detection methods, missing values, skewed numerical variables and "Unknown" labels. We investigate whether planet types and sizes vary with host-star class or discovery era, whether host-star metallicity differs between giant and rocky planets, how accurately available features predict planet radius, and whether a temperature-based habitable-zone label agrees with a stellar-flux estimate. Because the data are observational, observed patterns indicate statistical association rather than causation.

4. Literature Review / Background Study

Buchhave et al. showed that small planets form across a wide range of host-star metallicities, whereas giant planets occur preferentially around metal-rich stars [1]; Osborn and Bayliss confirmed this metallicity trend for hot Jupiters [2]. Mordasini et al. and Müller et al. characterised the planetary mass–radius relation and highlighted how composition and measurement precision shape its slope [3, 4]. Kopparapu et al. defined habitable-zone boundaries from stellar flux and climate limits, providing a more physically grounded benchmark than equilibrium temperature alone [5].

5. Methodology

5.1 Data Collection, Preprocessing and Tools

Loaded with pandas, nasa_exoplanet_intelligence.csv has 6,150 rows, 31 columns, zero duplicates and 6,346 missing cells (equilibrium temperature 25.4% missing, radius 0.8%, mass 0.5%).

"Unknown" strings were converted to NaN, discovery era was split into Pre-2015 vs 2015+, and right-skewed positive variables were log10-transformed; statistical tests used complete cases rather than imputed source values. A separate linear regression estimates missing equilibrium temperatures from host-star temperature, semi-major axis, and stellar radius; estimates are stored in a new column and do not overwrite observed values. Outliers were inspected via the IQR rule; five directly imaged orbits (>10^6 days) were excluded from clustering, and repeating H1–H4 without 49 controversial records left all decisions unchanged. Python libraries included pandas, NumPy, Matplotlib, Seaborn, SciPy and scikit-learn (seed = 42).

6. Implementation / Experimentation

After log10 transformation, planet mass and radius showed a strong positive correlation (r = 0.918, n = 6,076), while radius skewness dropped from 1.74 to 0.26. Transit accounted for 73.4% of discoveries, followed by radial velocity (19.2%). Mini-Neptunes were most frequent (2,148), followed by Gas Giants (1,734) and Super-Earths (1,185). Discoveries peaked in 2016 (1,496 planets; 2026 is a partial year). Median radius decreased across F, G, K and M stars (3.4, 2.8, 2.6 and 1.9 Earth radii), describing sample composition rather than direct causation.

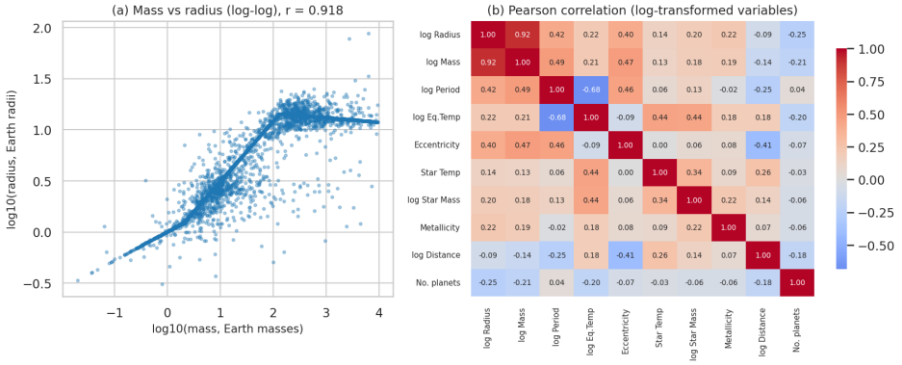


Figure 1. Log mass–radius relationship (r = 0.918) and Pearson correlation matrix across log-transformed variables.

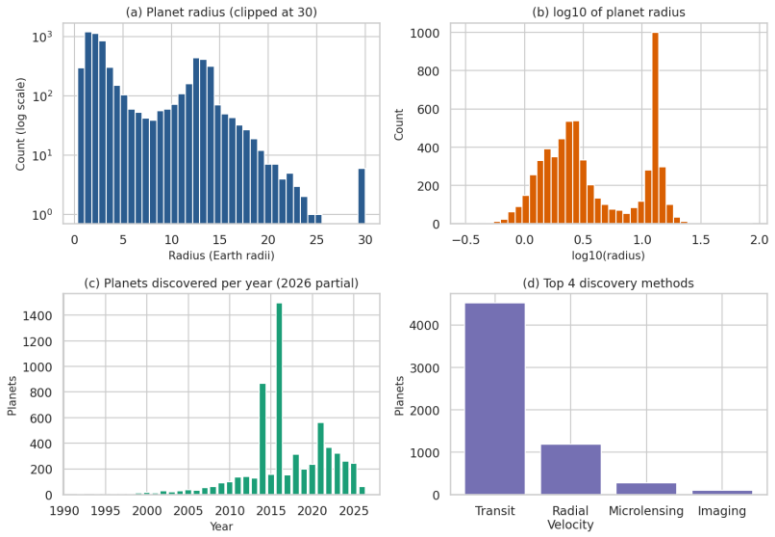


Figure 2. Distributions of planet radius, log10(radius), annual discoveries (2026 partial), and top discovery methods.

7. Results and Discussion

7.1 Confirmatory Data Analysis (CDA)

All six hypothesis tests were evaluated at $\alpha = 0.05$. Expected cell counts exceeded 5 in H1 and H5 (validating chi-square), and F/G/K/M log-radius standard deviations were similar in H3 (0.38–0.43). With thousands of rows, small p-values confirm statistical significance but must be read alongside effect sizes and plots.

Test	Null Hypothesis (H_0)	Result and Interpretation
H1: Chi-square	Planet type is independent of F/G/K/M host-star class.	$\chi^2(15) = 482.9$, $p = 2.37 \times 10^{-93}$. Reject H_0; distributions differ by class.
H2: Welch t-test	Mean host metallicity is equal for giant and rocky planets.	$t(3074) = 14.37$, $p = 2.24 \times 10^{-45}$. Giants' hosts have higher mean [Fe/H] (0.071 vs −0.028).
H3: One-way ANOVA	Mean log radius is equal across F/G/K/M hosts.	$F(3, 5761) = 58.44$, $p = 3.30 \times 10^{-37}$. At least one group mean differs.
H4: Welch t-test	Mean log radius is equal before 2015 and from 2015 onwards.	$t(3030) = 9.44$, $p = 7.01 \times 10^{-21}$. Geometric mean radius: 4.56 vs 3.53 Earth radii.
H5: Chi-square	Flux-based optimistic HZ membership is independent of F/G/K/M host class.	$\chi^2(3) = 75.4$, $p = 2.93 \times 10^{-16}$. HZ membership proportions differ by class.
H6: Two-way ANOVA	No star-class effect, era effect or interaction on log radius.	Star class $p = 8.82 \times 10^{-33}$; era $p = 1.01 \times 10^{-25}$; interaction $p = 0.00393$. All significant.

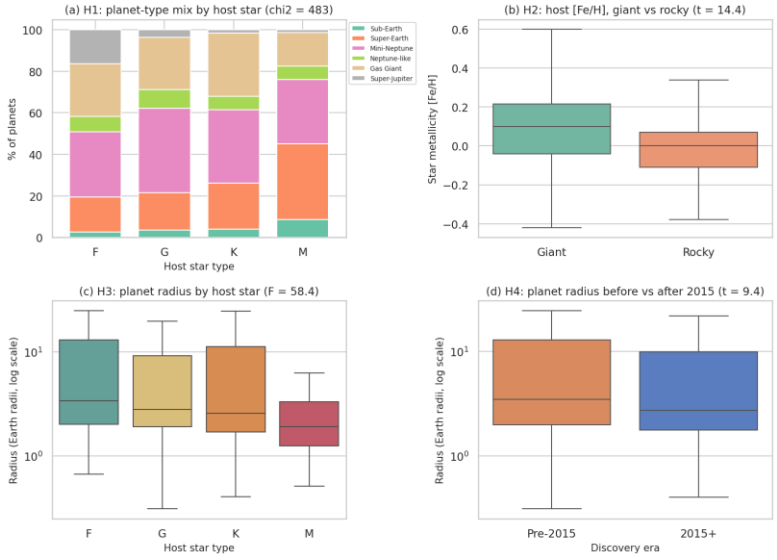


Figure 3. Visual comparisons for H1–H4: planet-type mix by star class, host [Fe/H] by group, and radius by star class and era.

7.2 Regression

Complete records were split 80/20 into training (4,395 rows) and test (1,099 rows) sets to predict log10(radius). Model A includes log(mass) (99.5% present, though some archive masses are empirical estimates from radius), whereas Model B excludes mass to isolate how much orbital period, stellar mass/temperature and metallicity explain.

Model	R^2	RMSE	MAE	Interpretation
A: with mass	0.844	0.155	0.106	Strong test fit; much of the fit reflects mass–radius association.
B: without mass	0.396	0.306	0.233	Other features explain less of the variation.

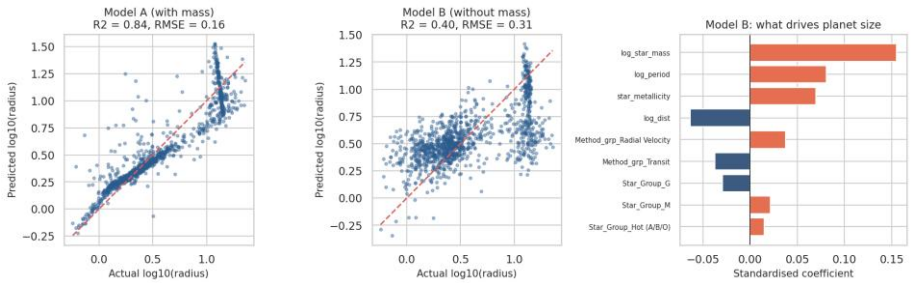


Figure 4. Held-out test predictions for Model A and Model B, alongside standardized coefficients for Model B.

7.2.1 Missing Equilibrium-Temperature Estimates

A separate linear regression was fitted on 4,570 planets with observed equilibrium temperature and complete predictor values. Using host-star temperature, semi-major axis, and stellar radius, it estimates 1,262 of the 1,563 missing temperatures; 301 records lack at least one predictor. Estimates are written to `predicted_equilibrium_temp_k`, leaving `equilibrium_temp_k` unchanged and keeping the estimates out of the hypothesis tests. The in-sample R^2 of 0.238 is a training fit, not independent validation. Estimates reach approximately 27,506 K, above the observed maximum of 4,050 K, so extreme values should be reviewed before interpretation.

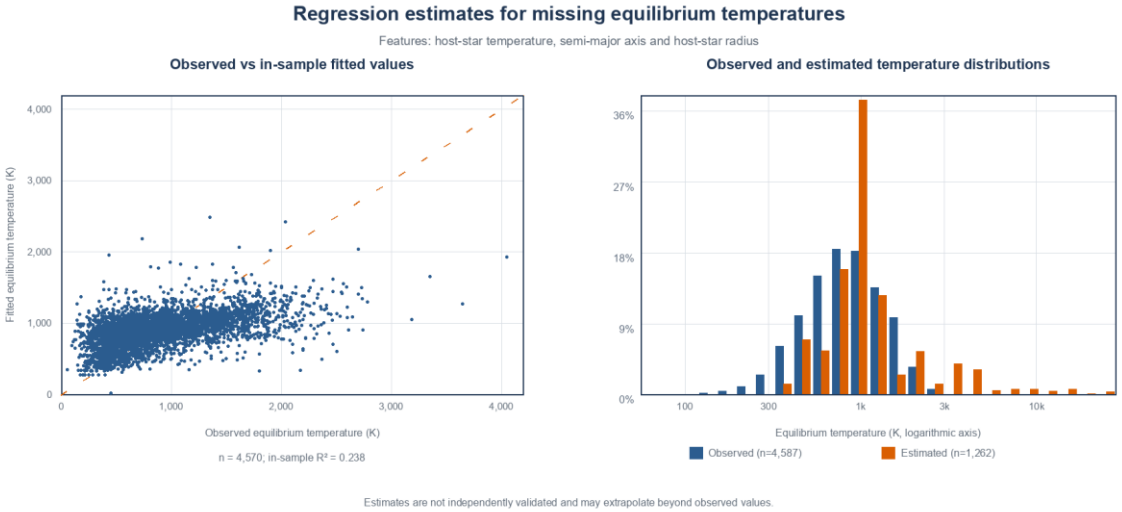


Figure 5. Added equilibrium-temperature regression diagnostic. Left: observed versus in-sample fitted values for 4,570 complete training records. Right: observed and estimated temperature distributions on a logarithmic axis; estimates are not independently validated.

7.3 Clustering and PCA

KMeans clustering on 4,500 complete records across six standardised features yielded silhouette scores of 0.373 (k = 3) and 0.372 (k = 2). Retaining k = 3 for exploration separates cool low-mass star planets, hot gas giants, and close-in sub-Neptunes/super-Earths across two PCA axes explaining 73.8% of total variance (46.0% + 27.9%).

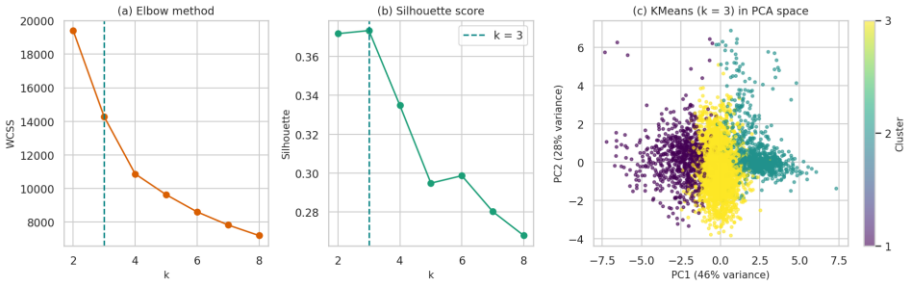


Figure 6. KMeans elbow curve, silhouette scores across k = 2..8, and the k = 3 solution projected into 2D PCA space.

7.4 Habitable-Zone Comparison

The dataset's habitable_zone_flag (200–400 K; 403 planets) was compared against Kopparapu et al. [5] stellar-flux boundaries across 5,788 valid systems (294 optimistic HZ; 172 conservative HZ). Only 109 planets matched both criteria, while 286 temperature-flagged worlds fell inside the inner greenhouse limit (too hot). Screening non-controversial 0.5–2 R⊕ worlds in the optimistic flux HZ shortlisted 43 candidates led by Wolf 1069 b, TRAPPIST-1 e and TOI-700 d (representing candidate screening rather than confirmed habitability).

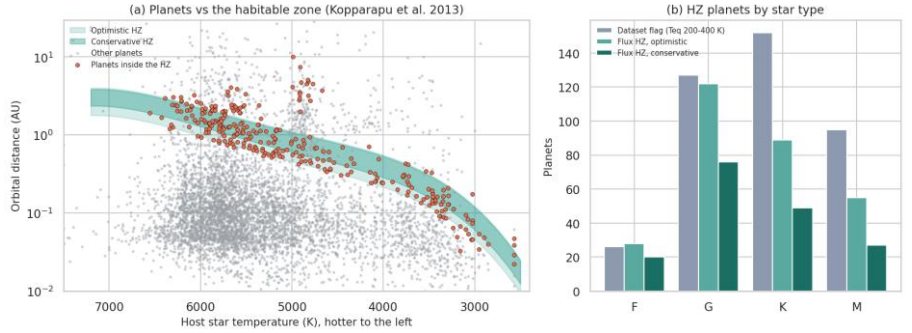


Figure 7. Kopparapu et al. (2013) flux-based habitable-zone boundaries and comparison with the dataset's temperature flag.

8. Conclusion and Future Work

This workflow audited 6,150 exoplanets, confirmed significant group differences (H1–H6), showed that radius prediction drops from $R^2 = 0.844$ (with mass) to 0.396 (without mass), identified three exploratory PCA/KMeans regimes, and demonstrated why stellar-flux HZ bounds improve upon a 200–400 K temperature cut. A separate, unvalidated temperature regression generated 1,262 estimates in a new column while retaining all observed temperatures. Future work should separate measured from estimated masses, report effect sizes and post-hoc tests, and incorporate atmospheric spectra for shortlisted HZ candidates.

9. Team Member's Contribution

DIVYAM— Data quality, EDA, and group testing: Reviewed missing values and data quality, prepared the cleaned dataset, and summarized key distributions and planet types. Also contributed to hypothesis testing on how planet types vary across host-star groups and implemented the regression estimates for missing equilibrium temperatures.

NIVED — Relationships, EDA, and regression analysis: Explored correlations and patterns between planet, orbit, and host-star properties. Contributed to hypothesis tests comparing planet groups and eras and built and evaluated the planet-radius regression models.

JYTOHIR — Habitable zones, clustering, and synthesis: Explored planet clusters and host-star patterns using clustering and PCA. Calculated and compared conservative and optimistic habitable-zone results with the catalog's temperature-based flag, contributed to interpreting the statistical findings, and checked consistency across the notebook, cleaned dataset, report, and README.

10. References

[1] L. A. Buchhave et al., "An abundance of small exoplanets around stars with a wide range of metallicities," Nature, vol. 486, pp. 375–377, 2012. doi: 10.1038/nature11121.

[2] A. Osborn and D. Bayliss, "Investigating the planet–metallicity correlation for hot Jupiters," MNRAS, vol. 491, no. 3, pp. 4481–4487, 2020. doi: 10.1093/mnras/stz3207.

[3] C. Mordasini et al., "Characterization of exoplanets from their formation – II. The planetary mass-radius relationship," A&A, vol. 547, A112, 2012. doi: 10.1051/0004-6361/201118464.

[4] S. Müller, J. Baron, R. Helled, F. Bouchy, and L. Parc, "The mass–radius relation of exoplanets revisited," A&A, vol. 686, A296, 2024. doi: 10.1051/0004-6361/202348690.

[5] R. K. Kopparapu et al., "Habitable zones around main-sequence stars: New estimates," ApJ, vol. 765, no. 2, article 131, 2013. doi: 10.1088/0004-637X/765/2/131.

[6] NASA Exoplanet Archive, "Acknowledging the NASA Exoplanet Archive in Publications," https://exoplanet.ipac.caltech.edu/docs/acknowledge.html (accessed 25 Oct. 2026).

11. Appendix

The accompanying Jupyter notebook, NASA_Exoplanets_IA4.ipynb, saves nine analysis figures in figures/ and exports exoplanets_cleaned.csv with the additional predicted_equilibrium_temp_k column. Place nasa_exoplanet_intelligence.csv in the notebook working directory and run the cells from top to bottom to reproduce the analysis. The working code (Both Jupyter Notebook and Python), Original Dataset is in the GitHub link attached below. And the Cleaned Dataset, having regressed values and Log Transformations in separate Columns is also in the GitHub Repository.

GITHUB LINK: -

https://github.com/jsnv25/Exoplanet-Analysis_IA4